More Problems and Solutions

UBCO MDS — DATA 582

Inference on Multiple Parameters

- To this point, we've
  1. introduced the Bayesian paradigm broadly;
  2. discussed point estimators and credible intervals;
  3. used sampling techniques to bypass difficult posterior distributions.

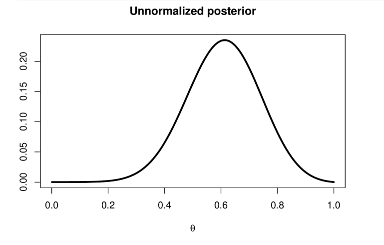
- In all cases we've discussed, we've been focused on the single parameter case. Specifically, either p from Bernoulli(p) or μ from $N(\mu, \sigma^2)$ with σ^2 known.

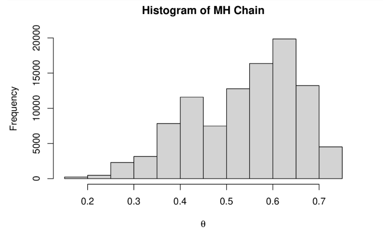
- Standard progression: what if σ^2 is unknown?

- There could be two special cases to consider here
  - we are also interested in inference on σ^2
  - we only care about inference on μ (σ^2 is then considered a nuisance parameter)

But first!

- Before we tackle multiple parameters in a Bayesian framework, let's do a little more troubleshooting on MCMC sampling.
- We showed some histograms resulting from an MCMC process, including some cases in our practice problems that looked suspicious...





But first!

- This begs an important question: namely, how will we know that our MCMC sample is useful for inferential purposes?
- For one parameter, this is not terribly tricky because we can visualize (like we just have). But in higher dimensions this may not be feasible.
- Keeping in mind that Bayesian inference is merely treating our parameter θ as a random variable, let's simplify things and just pretend we want to use MCMC to sample some random variable X.

Normal Distribution

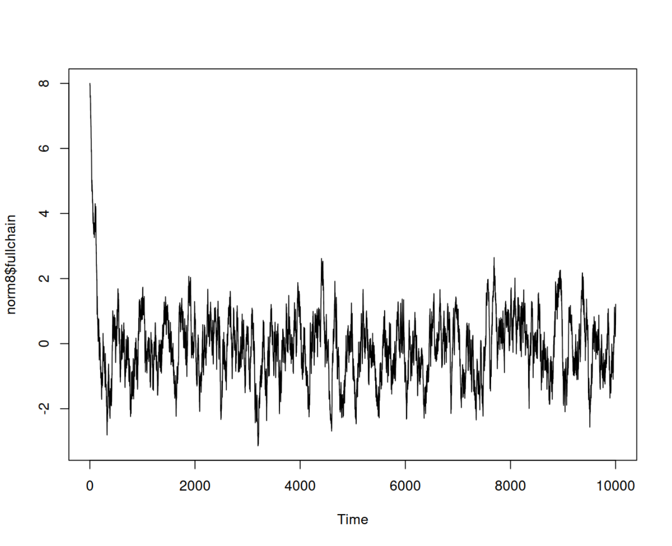
- Furthermore, let's assume $X \sim N(0, 1)$. We want to sample X, and we're going to pretend that's hard.
- We set up our M-H algorithm as
  1. Starting position $x^t = 8.0$ (bad start), generate a candidate value y via a proposal density $w(y \mid x^t) = x^t + Z$ where $Z \sim N(0, 0.2^2)$
  2. Calculate acceptance probability

$$a(y \mid X^t) = min\left(1, \frac{\phi(y)}{\phi(x^t)}\right)$$

  3. With probability a, set $X^{t+1} = y$, else set $x^{t+1} = x^t$
  4. Return to 1 until a set number of iterations are run.
- One good way to diagnose an MCMC chain is to visualize the chain of solutions as the sampling scheme progresses...these are called trace plots.

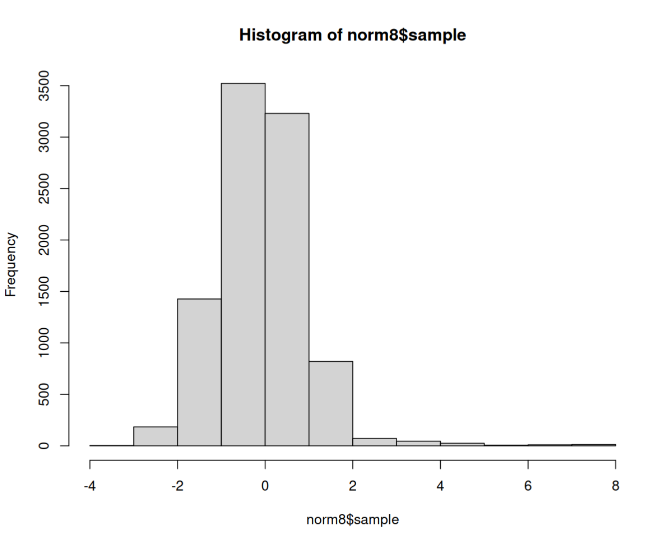
Normal Chain

- Do you notice any concerns?



Normal Sample Histogram

- Do you notice any concerns?

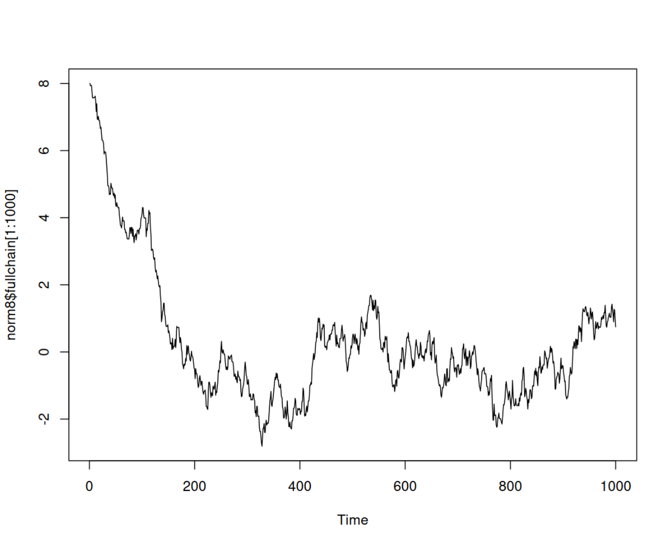


'Burn in'

- This is a typical MCMC problem when the initialization is far from the mode(s) of the target distribution.
- The period of time that the random walk takes until the it reaches the stationary distribution is often called the burn-in period.
- As we've seen, including this period in the observed sample can cause issues!

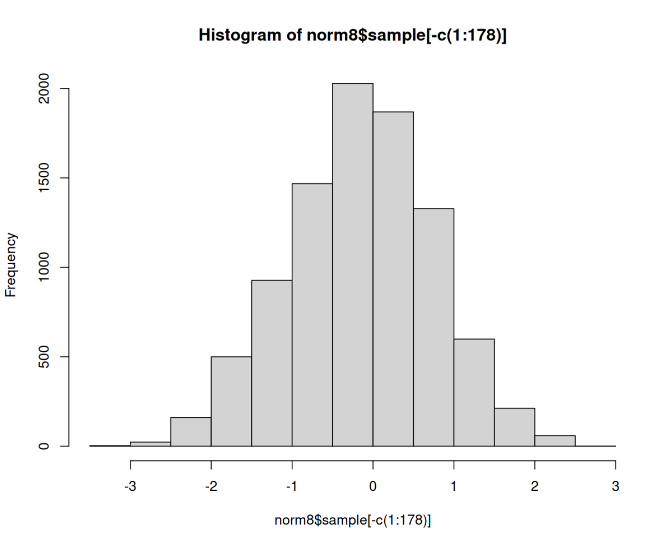
Normal Chain

- Zoomed in...



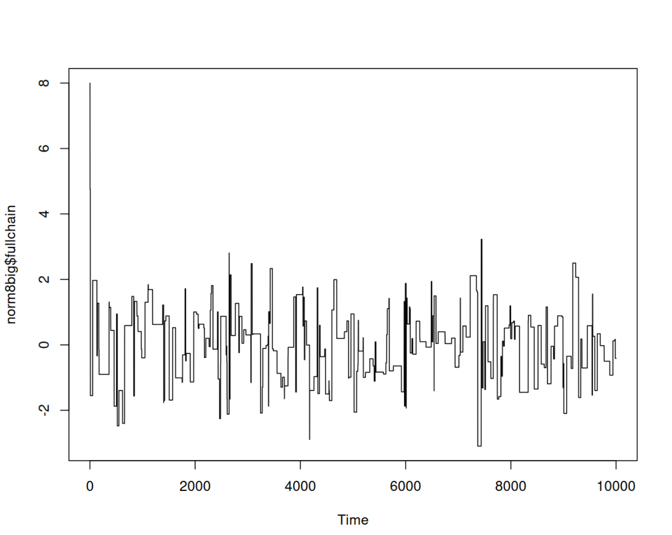
Normal Sample Histogram

- Discarding the first 200 samples...



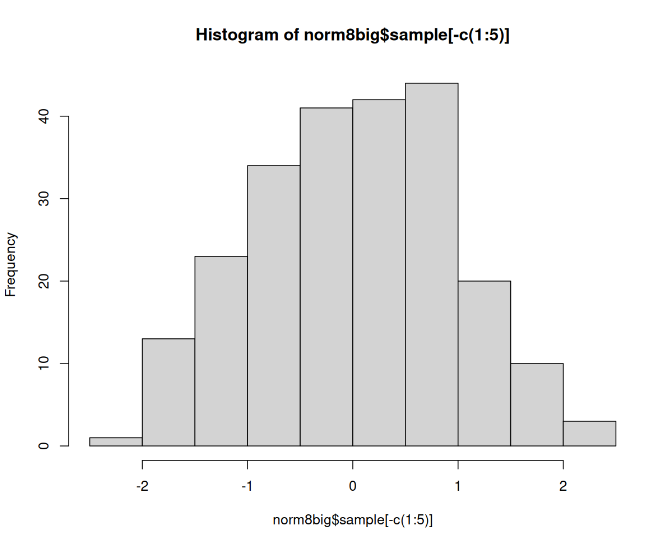
Normal Chain

- What if we greatly increase the variability of our proposal density $Z \sim N(0, 0.2^2)$ to $Z \sim N(0, 52^2)$. Concerns?



Normal Sample Histogram

- Discarding the first 100 samples for burn-in...concerns?

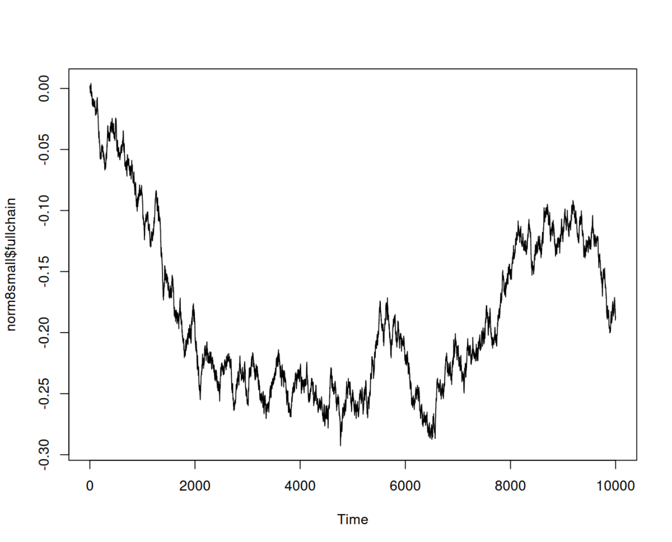


Too Many Rejections

- The proposal jumps are just too large and inefficient.
- Our sample size here is pretty small for the number of proposals, and it will take us too long to get a representative sample from this target distribution.

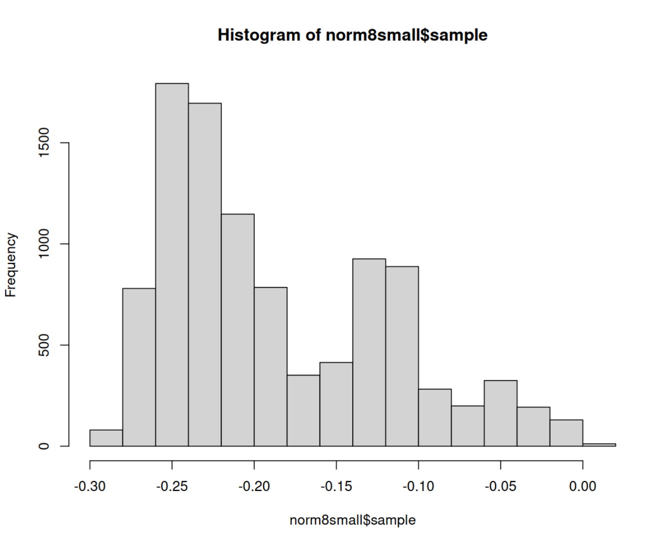
Normal Chain

- Suppose we start at $X = 0$ and use $Z \sim N(0, 0.002^2)$. Concerns?



Normal Sample Histogram

- Just for fun...concerns?



Too Small Steps

- The proposal jumps are just too small. We accept a large amount of proposals, but we're not exploring the domain of X well.
- Our sample size here is pretty small for the number of proposals, and it will take us too long to get a representative sample from this target distribution.

Summary

- Trace plots can help try to determine convergence to a stationary distribution.
- For simple-ish unimodal distributions, they will generally look like random error bouncing around a stationary point.
- If they look like step functions for large iteration counts, that's usually a sign that the rejection rate is too high.

Bivariate Distributions

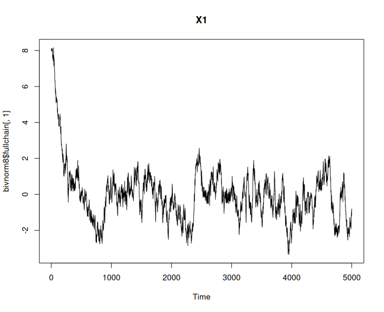
- Again, keep in mind the whole ‘parameters as random variables’ idea. We’ll continue to stick with ‘regular’ random variables for notation sake.
- Let’s suppose we want to sample (X_1, X_2) via a bivariate normal distribution (and again, pretend this is a challenge that we need MCMC for).
- Metropolis-Hastings can be applied to this problem in the same sort of way, so long as we have the (potentially unnormalized) joint distribution $f(X_1, X_2)$.

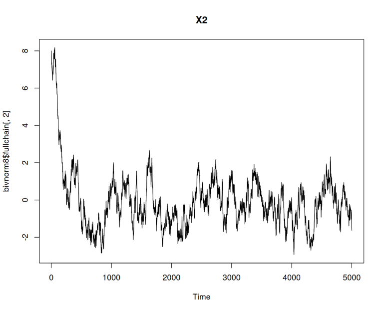
Bivariate Normal

- We now assume that $f(x_1, x_2) = f(\mathbf{x}) = \phi_2(\mathbf{x} \mid \mathbf{0}, \Sigma)$ where Σ has 0.5 on the off-diagonal and 1 on the diagonal.
- We set up our M-H algorithm as
  1. Starting position $\mathbf{x}^t = (8.0, 8.0)$ (bad start), generate a candidate value $\mathbf{y}$ via a proposal density $w(\mathbf{y} \mid \mathbf{x}^t) = \mathbf{x}^t + Z$ where $Z \sim MVN(\mathbf{0}, (0.2^2)I)$
  2. Calculate acceptance probability
  $$a(\mathbf{y} \mid \mathbf{x}^t) = min\left(1, \frac{\phi_2(\mathbf{y} \mid \mathbf{0}, \Sigma)}{\phi_2(\mathbf{x}^t \mid \mathbf{0}, \Sigma)}\right)$$
  3. With probability a, set $\mathbf{x}^{t+1} = \mathbf{y}$, else set $\mathbf{x}^{t+1} = \mathbf{x}^t$
  4. Return to 1 until a set number of iterations are run.
- Now we're generating a bivariate sample!

Two Trace Plots

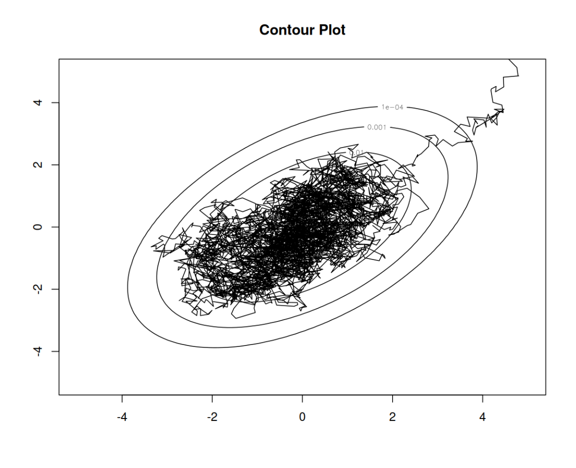
- Any concerns?





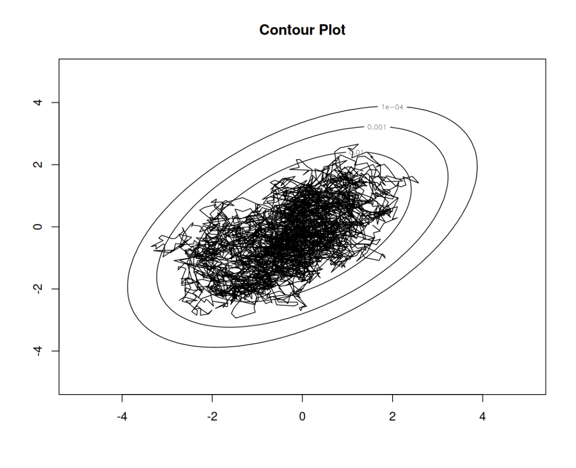
Chain on Contour

- We can monitor the chain in both dimensions simultaneously



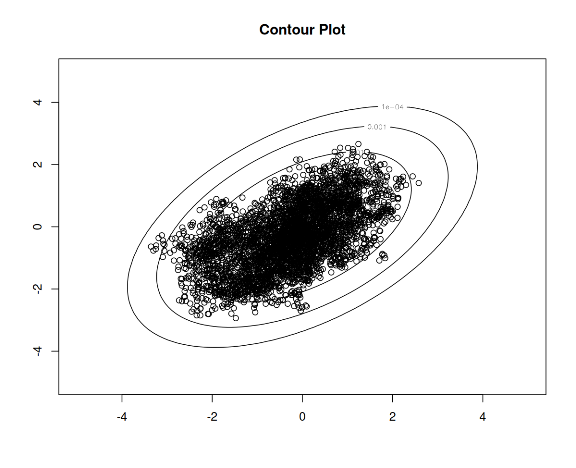
Chain on Contour

- Remove burn-in...



Sample on Contour

- And show sample...concerns?

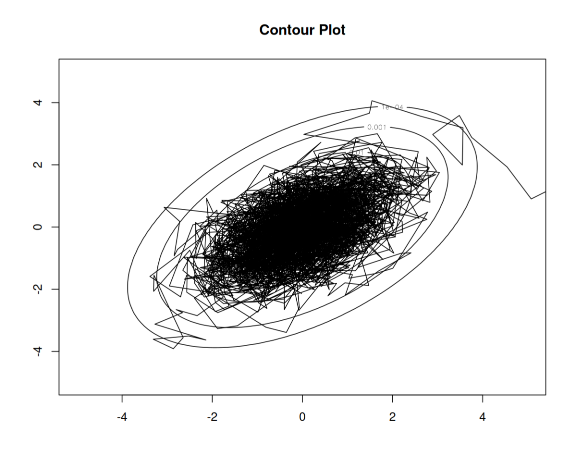


Too Dependent!

- While not as drastic as another case we've seen, there's some detectable local dependencies in this sample.
- We're accepting too many proposals and not exploring the space all that well.
- So we increase the standard deviation of the proposal distribution from 0.2 to 1.0...

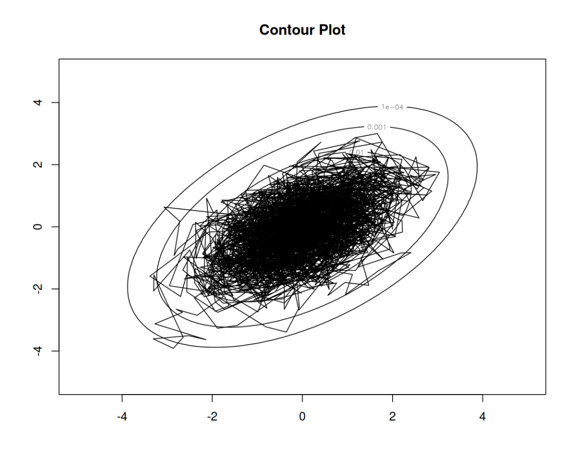
Chain on Contour

- We can monitor the chain in both dimensions simultaneously



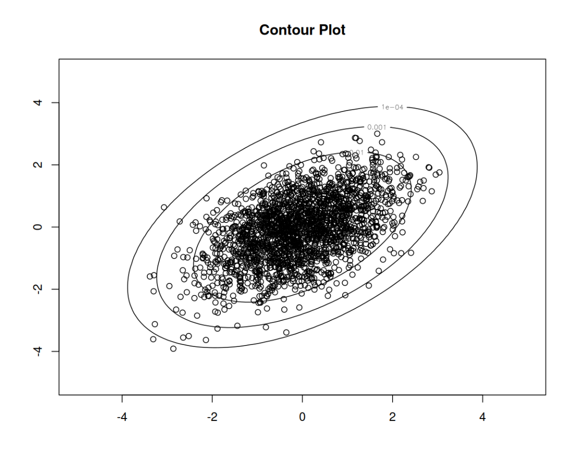
Chain on Contour

- Remove burn-in...



Sample on Contour

- And show sample...much better...



Conditional Distributions

- In some cases, the conditional distributions $f(x_1 \mid x_2)$ and $f(x_2 \mid x_1)$ may be easy to sample from.
- If so, there is another approach we can take advantage of: the Gibbs sampler.

Gibbs Sampler

- The basic idea is to walk through the MH algorithm but iterate back and forth between the two random variables.
- So with x_1^t we generate x_2^t. Then with x_2^t we generate x_1^{t+1}, etc.
- There are efficiency benefits to this approach. In fact, approaching sampling in this manner guarantees no rejections. Which means it is the optimal sampling approach, when available.
- But we can't always find conditional distributions...and we can also run into problems when the random variables are highly correlated.